

What's New and What's Next for Capella

THURSDAY, SEPT 10th, 2026



**Stéphane
LACRAMPE**
OBEO

Don't miss out the next events!



WEBINAR

with



Connect Capella to Your AI

Ecosystem by Stephane Lacrampe (Obeo)

Sept 24, 2026

Online

<https://mbse-capella.org/webinars.html>



Save the date! The event will take place December 1-3, 2026

Program announced on September 17

https://mbse-capella.org/capella_days.html

**ARCADIA AND
CAPELLA TRAINING**

Sept 28 - Oct 1 & 5-6, 2026

| 2:00-5:30 PM, Paris time (UTC+2)

Nov 16-23, 2026

| 2:00-5:30 PM, Paris time (UTC+2)

Dec 14-21, 2026

| 2:00-5:30 PM, Paris time (UTC+2)

<https://www.obeosoft.com/en/services/training/capella-training>





Stephane LACRAMPE
*Managing Director, North and
South America*
Obeo

Agenda

- **Introduction to Arcadia & Capella**
- The Capella Community
- What's New in Capella 7.1.0
- Capella Open-Source Add-ons
- Ecosystem Products
- Capella for SysON
- Q&A

© Copyright Obeo

The diagram illustrates the relationship between 'Need understanding' and 'Solution architectural design' through a series of hierarchical layers and components.

Need understanding:

- ViewPoints:** A hierarchy of components. The top layer includes A1, A2, and A3. The middle layer includes F1, F2, F3, F4, and F5. The bottom layer includes C1, C2, and C3.
- Reqs:** A block representing requirements, connected to the top layer of ViewPoints.

Solution architectural design:

- ViewPoints:** A hierarchy of components. The top layer includes F1, F2, F3, F4, F5, and F6. The middle layer includes C1, C2, C3, and C4. The bottom layer includes C1, C2, C3, and C4.
- Processors:** A block representing the processing layer, connected to the bottom layer of ViewPoints.

Legend:

- A: Operational activity
- F: Function
- C: Component

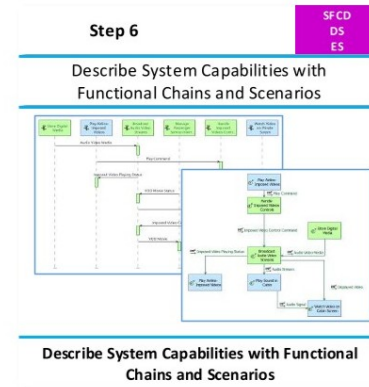
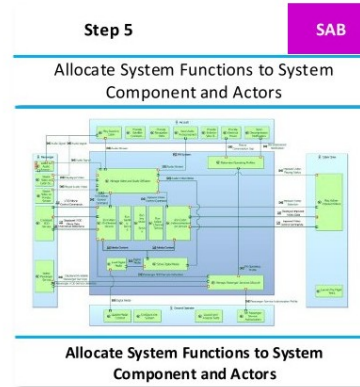
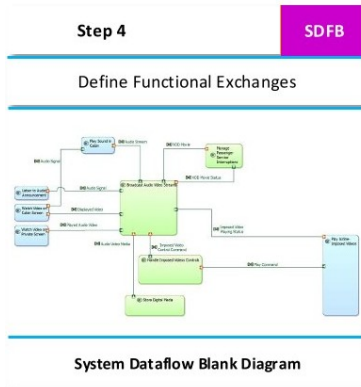
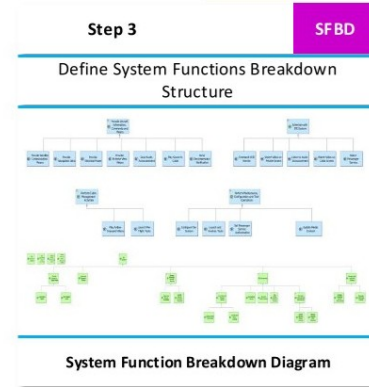
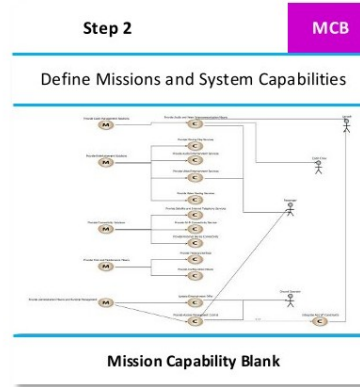
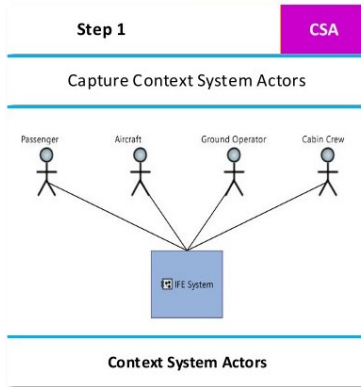
Flow and Dependencies:

- Red arrows indicate the flow of information and dependencies between components across the layers.
- Yellow arrows indicate specific functional or component relationships within the layers.
- Blue arrows indicate the flow of data or information between the layers.

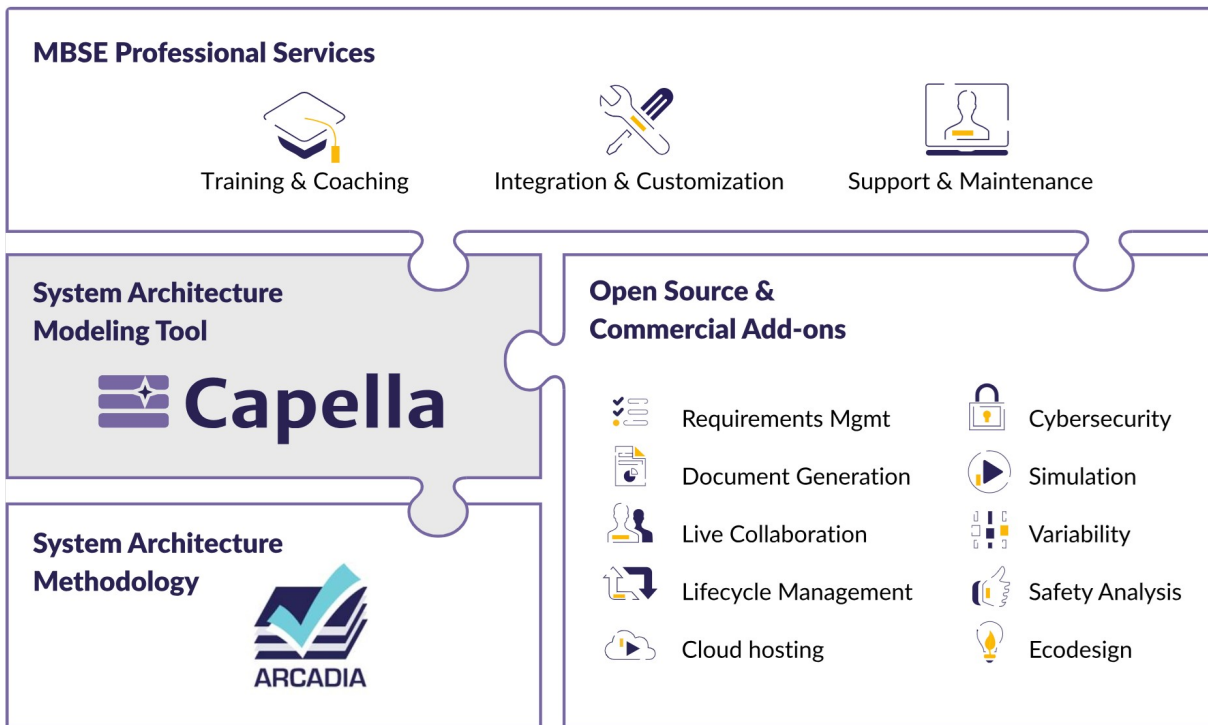
Physical Architecture
How the system will be developed and built



Arcadia/Capella Views



Capella ecosystem



Capella, the modeling tool implementing the Arcadia methodology, is the cornerstone of a comprehensive MBSE ecosystem.

Agenda

- Introduction to Arcadia & Capella
- **The Capella Community**
- What's New in Capella 7.1.0
- Capella Open-Source Add-ons
- Ecosystem Products
- Capella for SysON
- Q&A

A Thriving Community, Built Over a Decade

- 80+ webinars organized since 2017
- 9 editions of Capella Days (annual online conference)
- 180+ videos on the Capella YouTube channel
- 500+ LinkedIn posts/year
- Active forum: ~150 messages/month



Capella website content available in 8 languages

WORKBENCH ▾

TOOL FEATURES

GET STARTED

ADD-ONS

 ÜBERSICHT

 APERÇU

 RESUMEN

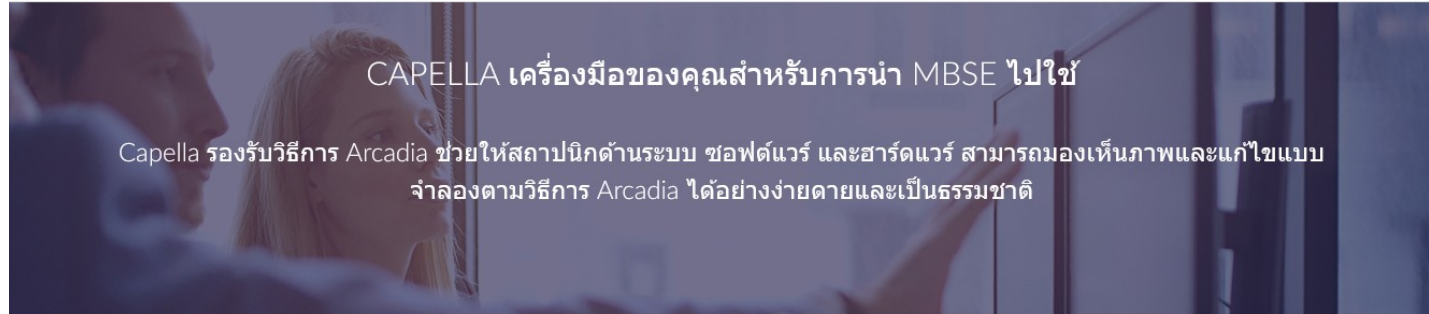
 VISÃO GERAL

 概要

 개요

 概述

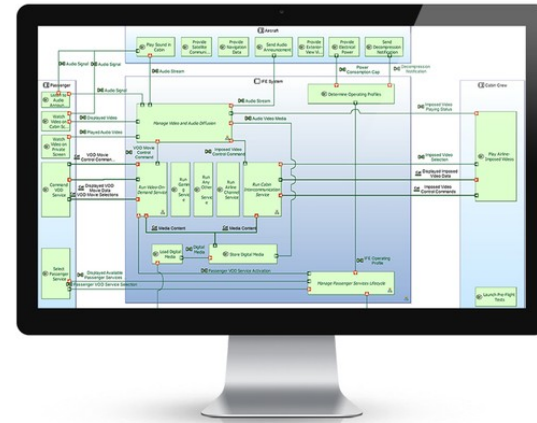
 ภาพรวม



Capella เป็นเครื่องมือ MBSE แบบโอเพนซอร์สที่สร้างแบบจำลองที่ให้แผนภาพหลากหลายประเภท โดยจัดเรียงตามมุมมองทางวิศวกรรม 5 มุมมองที่กำหนดไว้ใน Arcadia.

Capella มีฟังก์ชันการทำงานที่ช่วยให้สามารถปฏิบัติตามวิธีการของ Arcadia ได้อย่างราบรื่น เช่น การเข้าถึงกิจกรรมวิธีการที่มีบริการให้ใช้ทั้งหมด (Activity Explorer) และชุดเครื่องมือเร่งความเร็วในการสร้างแบบจำลองหรือเครื่องมือช่วยกระบวนการทางวิศวกรรมมากมาย ซึ่งช่วยอำนวยความสะดวกในการทำงานประจำวันของวิศวกรระบบ เพื่อให้พวกเขาสามารถมุ่งเน้นไปที่การออกแบบแทนที่จะต้องเสียเวลาไปกับการบำรุงรักษาแบบจำลอง

คุณสนใจของ Capella (ภาษาอังกฤษ) ➔



Capella at local and international conferences

- INCOSE International Workshop & International Symposium
- INCOSE AOSSEC, EMEA, GLRC, WSRC (US), SETE (Australia), Japan, Korea, India...
- ESA MBSE Conference, IEEE SysON, Siemens Realize Live, CSDM, AFIS...
- 160+ conferences since 2015
- Sponsoring, Capella booth, talks, tutorials...
- 1,500+ organizations using or having used Capella



2026 : a very active year

- 7 webinars since January
- Capella Days 2026 (Dec 1–3): Call for Proposals closed with strong response
 - talks from Rolls-Royce, Indra Space, Bluekei already preselected
- On the road: INCOSE IS (Yokohama), EMEASEC (Sweden), Dutch SE Days, TdSE, CSD&M, ASEC (UK), AeroSEC (India)...
- Growing local footprint: Capella meetups (Australia, Brazil...)

**WEBINAR**

Save the date
**Practice in COMAC to Conduct MBSA
in Avionics System Based on Capella**
中国商飞航电系统基于Capella的MBSA
应用实践

Tuesday, May 12th, 2026
4 PM-5 PM CST (Beijing)

**Hu
PO
COMAC****Renfei
XU
PGM**

Organized by
**OBEQ**

With the participation of
**COMAC PGM**



Agenda

- Introduction to Arcadia & Capella
- The Capella Community
- **What's New in Capella 7.1.0**
- Capella Open-Source Add-ons
- Ecosystem Products
- Capella for SysON
- Q&A

Capella 7.1.0: Stronger foundations



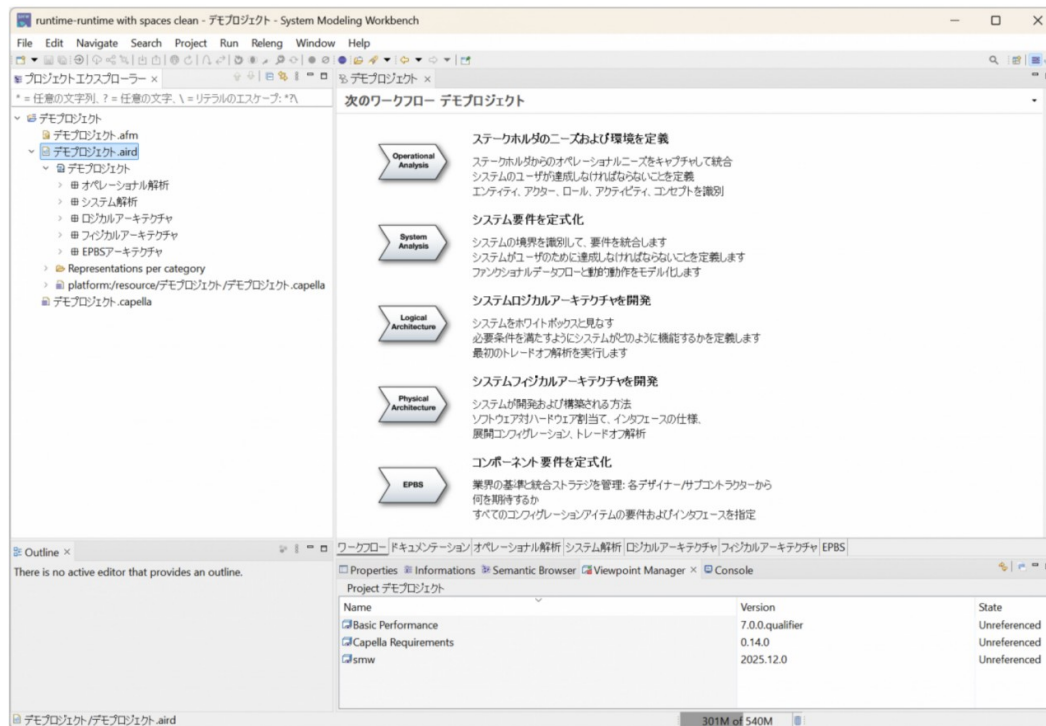
Building the future of Capella starts with strengthening its foundations: a more robust, secure and easier to maintain platform over the long term.

- **From Java 17 to Java 21 (more recent and sustainable version).**
- **CyberSecurity focus: nine CVEs addressed through dependency updates and removals, including:**
 - Removal of Mylyn features
 - Migration to reload4j 1.2.26
 - Migration to Guava 33
 - Migration to SLF4J API and Logback 1.5.32
 - Migration to EGit/JGit 6.6.1



Capella 7.1.0: Internationalization

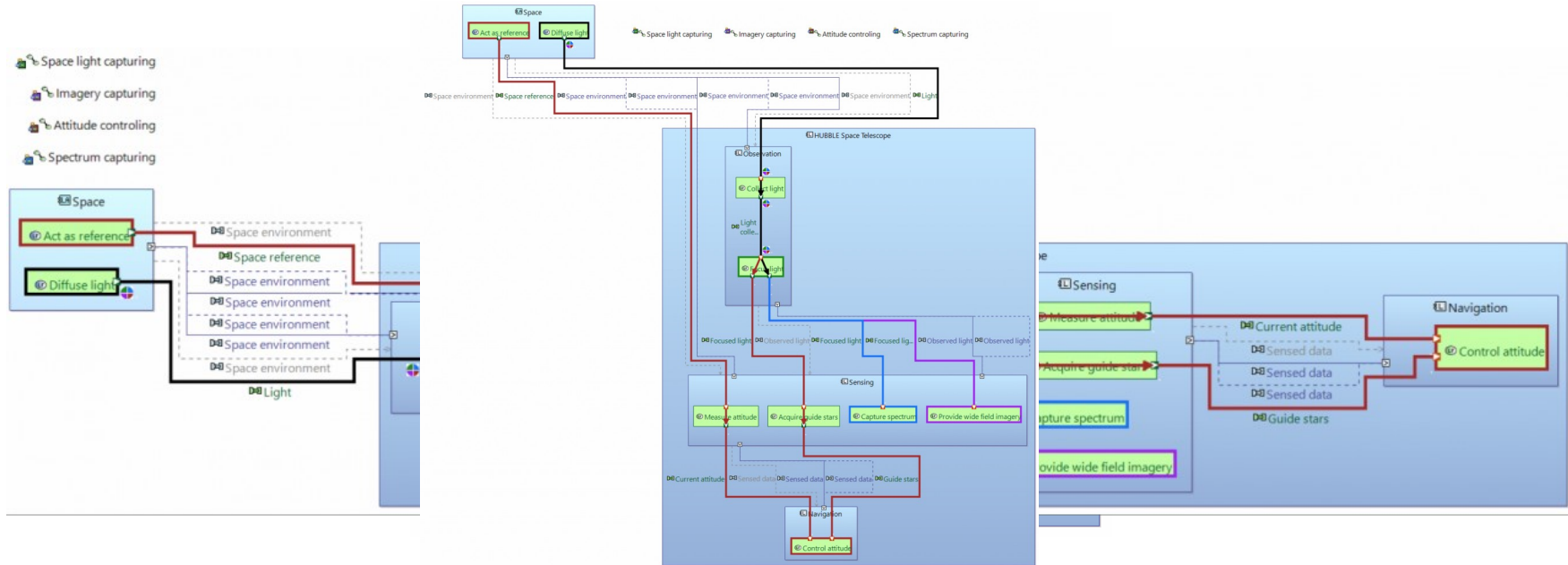
- The platform now provides the foundations needed to translate Capella's user interface into different languages.



Capella does not ship with translations out of the box yet: translations need to be provided separately.

Capella 7.1.0: Enhanced Layouting

- Diagram Layouts: probably the number one pain point in Capella
- Capella 7.1.0 includes a first integration of the Eclipse Layout Kernel (ELK) add-on, giving you access to a variety of automatic layout algorithms (tested on xAB diagrams)



Capella 7.1.0: Other improvements

- **Other improvements**
 - Project migration through APIs
 - REC/RPL fixes
 - Color disposal issues on Linux



Obeo commitment to Capella

- Obeo has been present **from the start**, supporting **Thales** in developing Capella.
- Our role has expanded since Capella was put in **open-source** in 2015
 - Actively supported the **international development** and **adoption** of Arcadia and Capella, establishing it as **leading reference** in the MBSE landscape
 - Actively maintained **key technologies** that support Capella, including Eclipse Sirius, EMF Services, GMF Runtime, and other Eclipse technologies
- This Capella 7.1.0 is the beginning of an **extended involvement** of Obeo in Capella
 - **New release cycle** (2026.X, 2027.X...) with 3 to 4 releases per year
 - New committers and project **co-leadership**
 - Extended direct **investments** and **contributions** to Capella
 - And other exciting topics...

What's Next for Capella 2026.11.0 ?

- We are ramping up our capability to release Capella and all the underlying components
- Significant Eclipse Platform upgrade (from 2023.03 to 2025.09)
 - Upgrade on all Capella dependencies
 - Improved OS compatibility and decreased CVE exposure
- Activation of the new version numbering (CalVer)
- Various bug fixes and improvements
- ELK integration and possible improvements



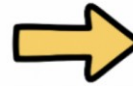
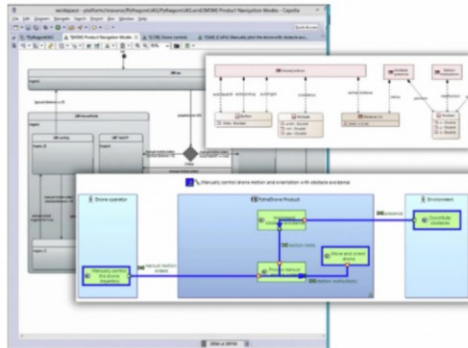
Agenda

- Introduction to Arcadia & Capella
- The Capella Community
- What's New in Capella 7.1.0
- **Capella Open-Source Add-ons**
- Ecosystem Products
- Capella for SysON
- Q&A

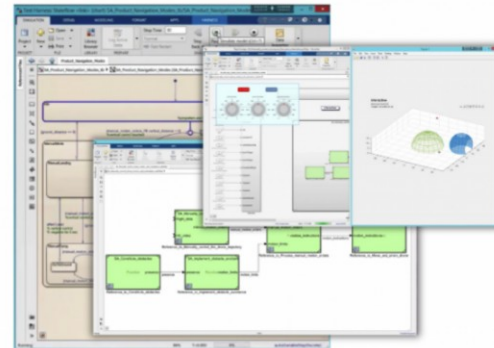
Capella Simulink® Connector

- **New** Open-Source add-on
 - **Developed by Thales** and open-sourced by Obeo
<https://github.com/eclipse-capella/capella-simulink-connector/releases/tag/0.2.0>
 - Integrate Capella state machines and functional chains with Simulink® for simulation
- Upcoming: Installation documentation and webinar presentation

 **Capella**

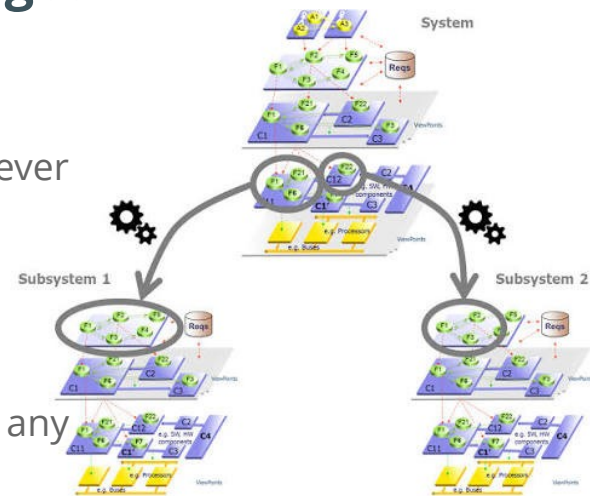


SIMULINK®



System to SubSystem transition: Safer Perimeter changes

- **Before**
 - Transition **assumed** the subsystem's component list (its "perimeter") never changed
 - Changing it often led to **model corruption**
- **Now**
 - Component selection is reviewed in the **transformation dialog** before any change is applied
 - Elements are **matched by role**, not technical ID → existing elements are updated, not destroyed and recreated
 - Impact on other model elements and diagrams is minimized
 - **Cancel button fixed:** no longer triggers unwanted modifications



Available in the latest version of the add-on and funded by Rolls-Royce



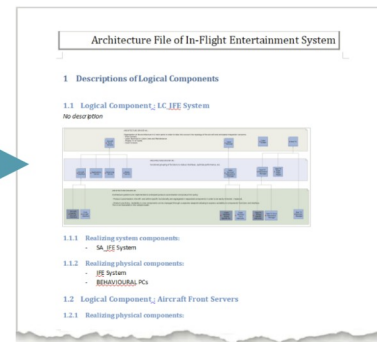
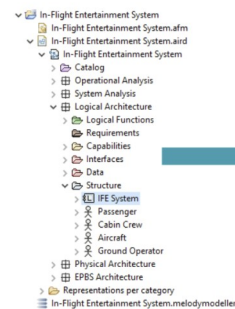
M2Doc: Keeping Word Generation Up to Date

- Recent evolutions

- Office 365 support & plain-text instructions (since M2Doc 4.0.0) -> better compatibility, easier to inspect templates
- Continuous improvements to HTML-to-Word integration (lists, tables, images)
- Section & page numbering via bookmark references for **accurate cross-references** in generated documents
- New: call a Semantic Browser query directly from a template : ***myLogicalFunction.getSBQuery('Functional Exchanges')***

- Looking ahead

- M2Doc 5.x is based on AQL 8 — adoption tied to Capella's own move to AQL 8
- Roadmap mainly driven by **customer needs**



Python for Capella — Scripting Made More Powerful

- **Recent evolutions (1.4.0)**

- New Diagram API: create diagrams, generate mappings, customize styles, by script
→ Demo
- Broader API coverage: fixes across Capella core, PVMT and Requirements add-ons
- Better packaging: core API and sample scripts now separate, more portable
- Python 3.12 compatibility

- **Looking ahead**

- Acceleo-based generator opens the door to auto-generating APIs for additional Capella add-ons

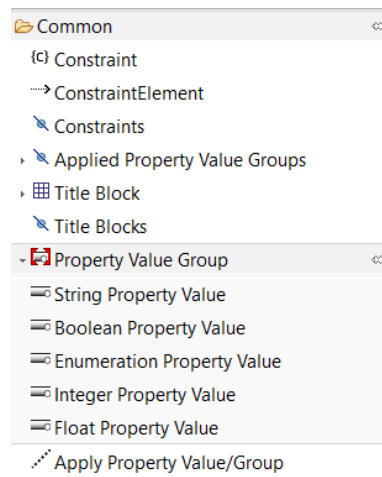
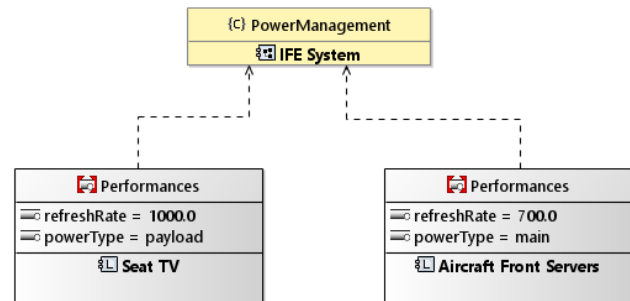
Capella Traceable Properties Values Edition

• What it does

- Makes non-functional requirements (mass, cost,...) easier to create, apply, visualize and trace in Capella
- Built on the Holistic Requirements Model: functional requirements say what the system does, non-functional requirements constrain how well

• How it works

- No new metamodel elements : reuses existing Capella concepts: Property Values, Property Value Groups, Applied Property Value Groups, Generic Trace links
- Adds usability: new palette tools + new diagram types on top of these existing concepts
- Property Value support directly on existing Capella diagrams + dedicated traceability diagrams for non-functional requirements
- Open-source (EPL), Available by end of November



Non-Functional Requirement Creation & Traceability using Property Values



Joshua Wedgwood
Rolls-Royce



Agenda

- Introduction to Arcadia & Capella
- The Capella Community
- What's New in Capella 7.1.0
- Capella Open-Source Add-ons
- **Ecosystem Products**
- Capella for SysON
- Q&A

Team for Capella



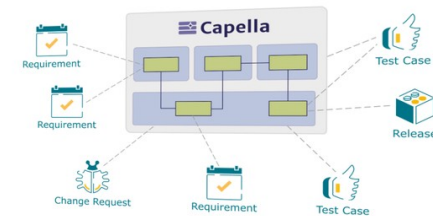
- **From 6.1.0 to 7.0.1_202601**

- **Authentication overhaul:** OpenID Connect rework, Authorization Code + PKCE, BCrypt-encrypted credentials, secrets via environment variables
- **Semantic Browser performance:** refresh time cut from ~1 min to **under 1 sec** on large connected projects
- **Server robustness:** CDO revision cache memory leak fixed, continuous Jetty/Jenkins CVE cleanup
- More reliable Import/Export wizards (special characters, **library detection**, unshared projects)

- **7.1.0 (latest) : Secure by Default**

- WSS/HTTPS now the **default connection mode**
- New: **per-repository capabilities** : control which Capella add-ons are authorized, per project
- **Extended OIDC:** opaque access token support (introspection)
- Aligned with Capella 7.1.0 foundations: Java 21, updated dependencies

Publication for Capella



- **Recent (2026.5 – 2026.7)**

- Broader ALM coverage: Jama Connect, Jira Cloud (via SodusWillert)
- New Organizations & Server Admin role for scalable access management
- Foundations for version control: history, baselines, previews
- Diagram images now published as Polarion work items

- **In progress**

- Bringing Capella–CodeBeamer integration to full traceability feature set
- Decorators in the online explorer (e.g., surfacing Filtering add-on filters)
- API access to object history, easing integration into customers' software landscape

- **Looking ahead**

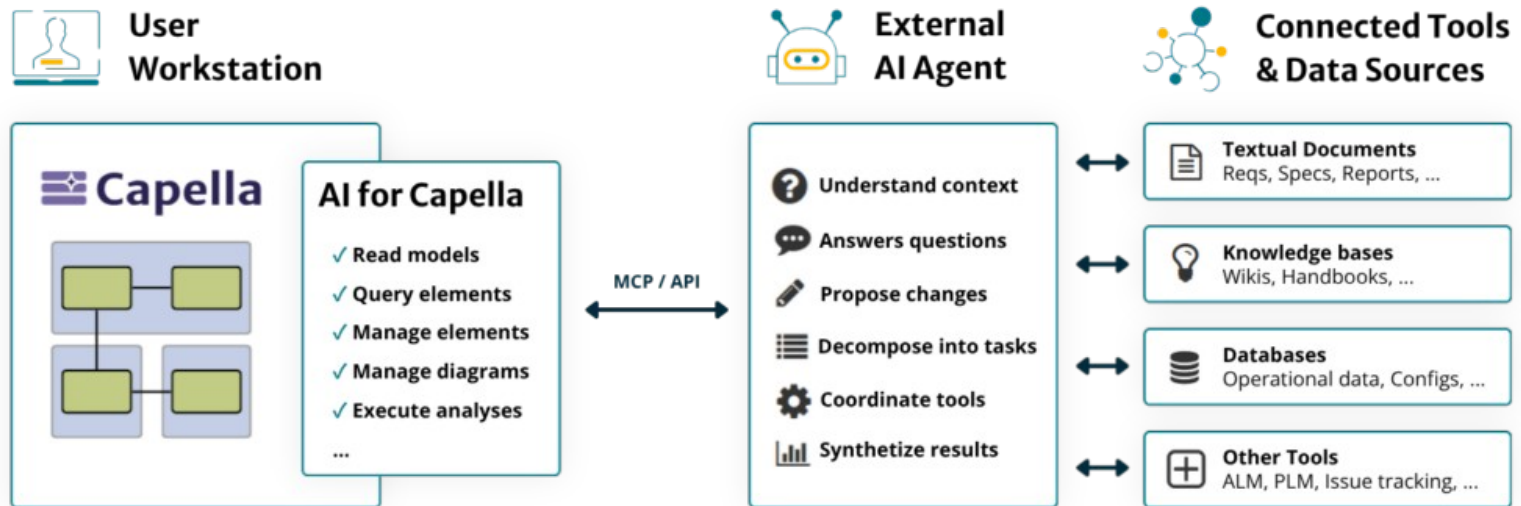
- AI-powered natural language querying of models
- Discussions on model elements

System Modeling Workbench (SMW)

- **2606**
 - Now based on Capella 7.1.0, requires Java 21
 - Key fix: **concurrency issue** in "update" that could corrupt models
 - Focus: **security & robustness**
- **What's next (2612)**
 - Eclipse 2025.09 upgrade & removal of legacy Chromium browser (both **CVE-driven**)
 - Exploring: rich-text images, SMW subsystem reuse from Teamcenter
 - New: multiple language support (installer-configurable), building on Capella 7.1's i18n foundations
 - Packaging the **EDA VSE add-on**

AI for Capella: Capella as an Agent-Ready Platform

- Rather than embedding one built-in assistant, AI for Capella **connects Capella to your AI ecosystem** : Claude, Codex, Mistral, a local model, or any agent platform
- **Controlled**, traceable interactions : **engineers stay in the loop**
- **Enterprise-ready** add-on for Capella Desktop



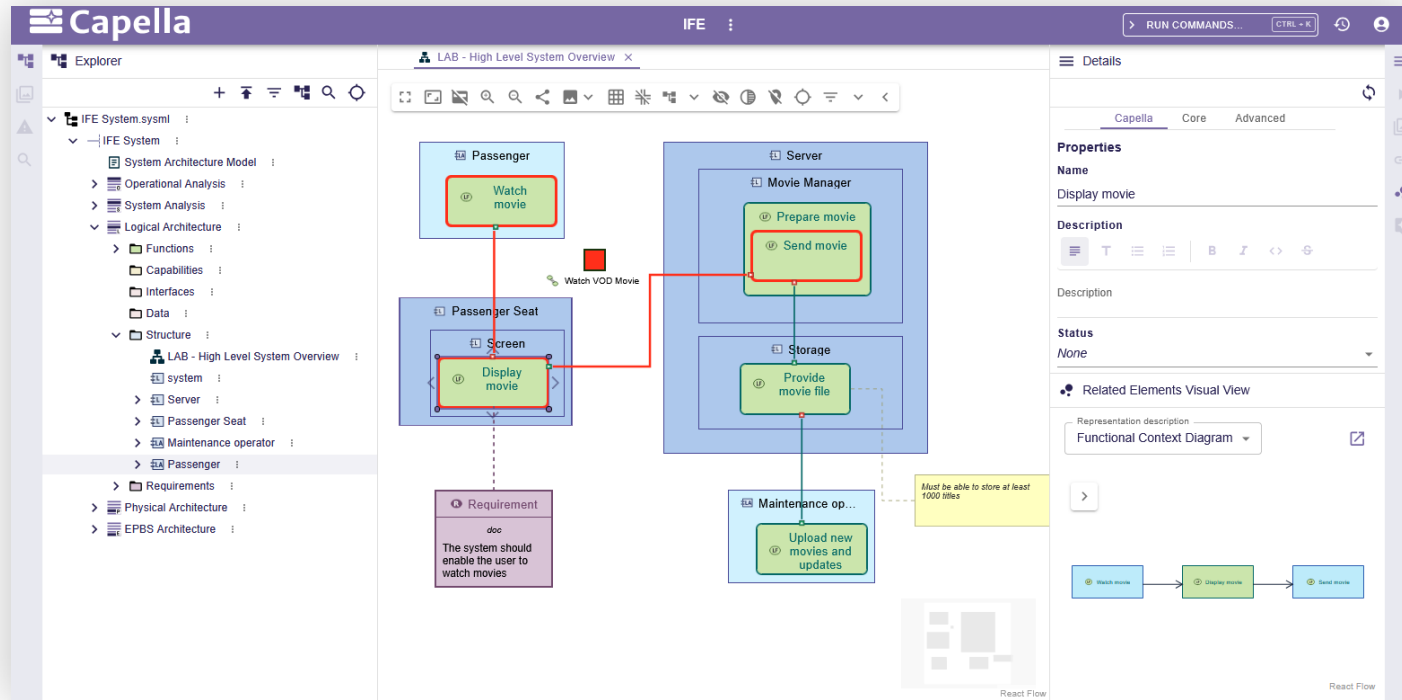
Other add-ons



Agenda

- Introduction to Arcadia & Capella
- The Capella Community
- What's New in Capella 7.1.0
- **Capella Add-ons**
- Ecosystem Products
- **Capella for SysON**
- Q&A

Capella for SysON: Towards the Next Generation of Capella



A new open-source initiative exploring the convergence of Capella, Arcadia, SysML v2, and web-based modeling

Introducing SysML v2

- Obeo is an **OMG member**
 - Part of the **SysML v2 Submission Team (SST)**, and actively contributes to the definition of SysML v2
- **Key evolutions / advantages**
 - New OMG standard, formally approved in **2025**
 - Built on KerML foundations: improved precision, expressiveness, and consistency
 - Native **textual** + **graphical** notation
 - Standardized API : positioning SysML v2 as the future **interoperability** standard across MBSE tools
- **The challenge**
 - SysML v2 is a **rich, powerful**, and **complex** general-purpose language
 - Its **level of abstraction** is **unchanged** from v1: it tells you how to express a model, not how to structure your systems engineering approach



**A structured method like Arcadia remains just as relevant,
and just as necessary, on top of SysML v2**

Sirius Web and SysON



- **Sirius Web**

- Cloud/low-code evolution of Eclipse Sirius: a low-code platform for building graphical modeling tools
- Modeling directly in the browser, built on modern web tech (React, Spring, GraphQL...)

- **SysON**

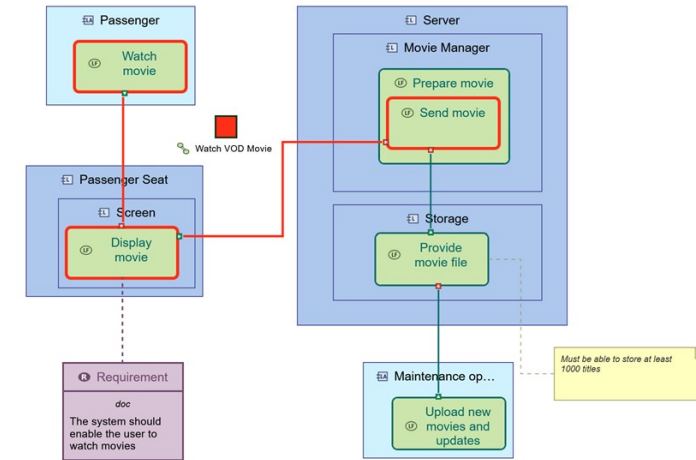


- Leading open-source, web-based SysML v2 editor, built on Sirius Web
- Co-developed by Obeo & CEA List, hosted at the Eclipse Foundation
- Flexible by design: customizable for domain-specific methodologies, without forking
- Already proven on real methodology extensions

**This is precisely what makes it possible to bring Arcadia
into a SysML v2, web-based environment**

What is Capella for SysON?

- **The idea**
 - Brings the Arcadia approach to SysML v2/SysON
 - Lets engineers model with Arcadia concepts, natively expressed in SysML v2
- **Three complementary open-source building blocks**
 - **Arcadia SysML v2 Library** : formalizes Arcadia concepts on SysML v2/KerML foundations (the semantic layer)
 - **Capella for SysON** : brings it all into SysON, with Arcadia-specific, Capella-inspired graphical editors
 - **Capella SysML v2 Bridge** : exposes existing Capella models in SysML v2

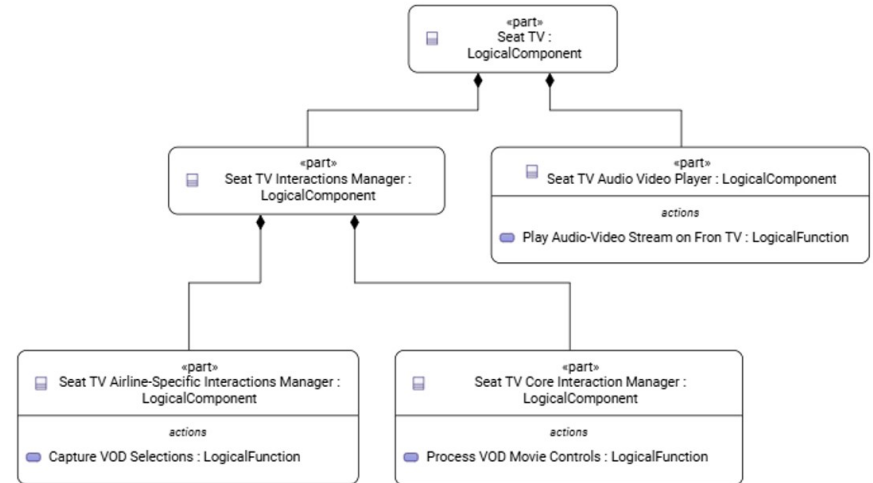


The Arcadia SysML v2 Library

- **The approach**

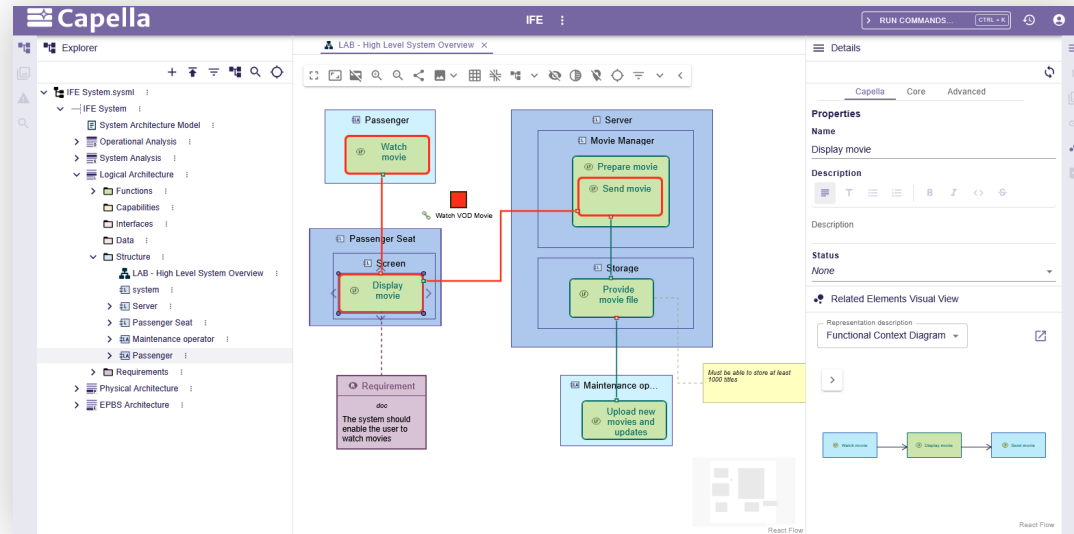
- Priority #1: **interoperability**. Models should stay as close as possible to "natural" SysML v2.
- A **minimal, essential semantic layer**: only the core concepts that matter (Functions, Components, Exchanges...)
- **Generic concepts** (Function, Component, Port) rather than perspective-specific ones (Logical/Physical Function): easier tooling, easier reuse across perspectives
- Based on **15+ years of Arcadia/Capella usage**: focus on what the community actually uses, not the full technical metamodel

```
part def Component :> ArcadiaElement {  
  private doc /* A generic Arcadia component, used as  
  attribute isActor : Boolean default false;  
  attribute isHuman : Boolean default false;  
  alias allocatedFunctions for performedActions;  
  alias subComponents for subparts;  
}  
  
port def ComponentPort :> ArcadiaElement {  
  private doc /* A generic Arcadia Component Port,  
  attribute portDirection : FeatureDirectionKind;  
  ref item allocatedFunctionPorts: ExchangeItem [*];  
}
```



Capella for SysON: A First Look

- What it brings
 - Capella-inspired **graphical editors**,
 - built on top of SysML v2 elements specialized with the Arcadia library
 - Same **modeling experience** and vocabulary Capella users already know, in a **browser**



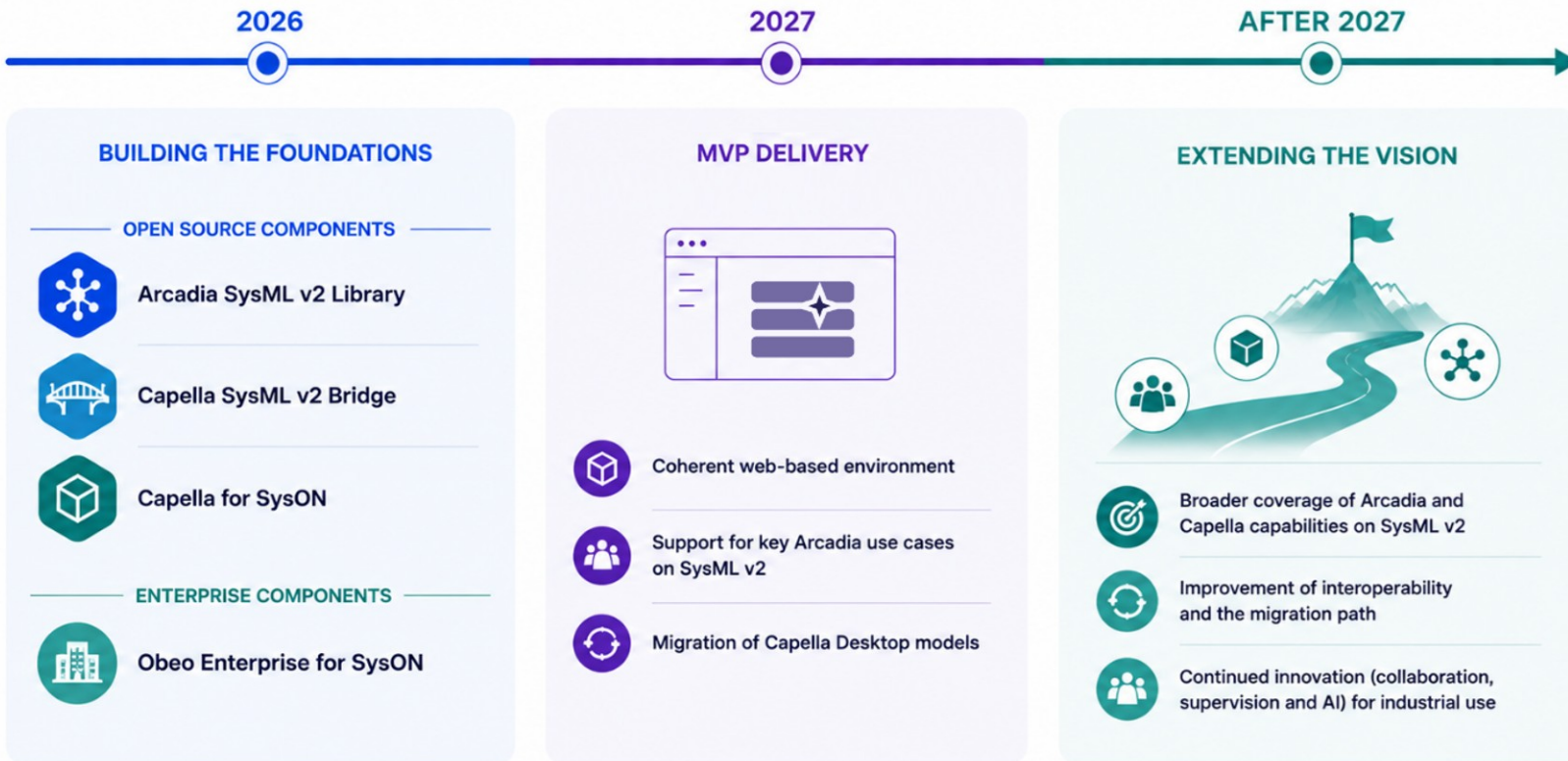
The Ambition: Building the Next Generation of Capella

- **A credible, progressive path toward a Capella environment :**
 - **Web-based** : easier deployment and collaboration in modern engineering organizations
 - **Compatible with SysML v2** : connecting Arcadia-based engineering to the emerging standard and its ecosystem
 - **Built for collaboration** : leveraging SysON's architecture: REST APIs, textual syntax
 - **Grounded in Arcadia** : preserving the methodological approach that is central to Capella
- **But not an overnight replacement**
 - **Capella is mature**: hundreds of wizards, helpers, and model transformations refined over 15+ years
 - Building the next generation means **progressively** bringing that value to a new foundation, not just reproducing diagrams in a browser

 <https://github.com/eclipse-capella/capella-for-syson>

Roadmap

Towards the Next Generation of Capella



The Capella SysML v2 Bridge: A Path Forward, Today

- **What it does**

- Exposes **existing Capella Desktop models** in SysML v2 : no need to leave Capella Desktop
- SysML v2 closed many gaps of v1 by moving **closer to Arcadia's concepts** : the transformation is far more natural than a v1 transformation would have been

- **Two use cases**

- A **migration path** toward SysML v2 / Capella for SysON
- Immediate **interoperability**: connect today's Capella models to the SysML v2 ecosystem (simulation, analysis...), even before Capella for SysON is mature

 <https://github.com/eclipse-capella/capella-sysmlv2-bridge>

Keep Modeling, and Help Us Shape and Build What's Next

- **If you use Capella today, keep using Capella**
 - Capella Desktop remains the **mature solution** for **operational** Arcadia projects : Capella for SysON is still **early-stage**
 - The path forward, in two steps: (1) expose Capella models in SysML v2 today via the Bridge, (2) migrate to Capella for SysON as it matures
- **Join the initiative**
 - Share your use cases, experiment, give **feedback**
 - **Co-fund** specific developments if this is **strategic for your organization**

**The future is not Capella or SysML v2.
It's Capella, with SysML v2.**

<https://blog.obeosoft.com/capella-for-syson>

Wrap-up

- Capella 7.1.0 is **stronger**, more **secure**, and more international — and it's just the beginning of a **bigger commitment**
- Obeo is investing **more**: 3 to 4 releases a year, new committers, extended contributions
- The ecosystem is thriving: new add-ons, real industrial contributions , a growing global community
- Capella for SysON: our path to the future



Thanks for listening!

Do you have any questions?

Let's keep on the discussion on the forum after the webinar:

<https://forum.mbse-capella.org/>

Don't miss out the next events!



WEBINAR

with



Connect Capella to Your AI

Ecosystem by Stephane Lacrampe (Obeo)

Sept 24, 2026

Online

<https://mbse-capella.org/webinars.html>



Save the date! The event will take place December 1-3, 2026

Program announced on September 17

https://mbse-capella.org/capella_days.html

**ARCADIA AND
CAPELLA TRAINING**

Sept 28 - Oct 1 & 5-6, 2026

| 2:00-5:30 PM, Paris time (UTC+2)

Nov 16-23, 2026

| 2:00-5:30 PM, Paris time (UTC+2)

Dec 14-21, 2026

| 2:00-5:30 PM, Paris time (UTC+2)

<https://www.obeosoft.com/en/services/training/capella-training>



Stay Tuned!



mbse-capella.org/capella_days_2026.html

mbse-capella.org/webinars.html



Good morning, good afternoon, or good evening, depending on where you're joining from — thank you all for being here.

Today's session is called 'What's New and What's Next for Capella.' As the title suggests, we'll cover two things: what's actually shipped recently — new versions, new add-ons, new capabilities — and where we're headed next, including some truly exciting news about the future of Capella that I'm looking forward to sharing with you today.

Before we dive in, a couple of things you shouldn't miss.

First, a quick heads-up on what's coming right after this.

On September 24th, I'll be hosting another webinar: 'Connect Capella to Your AI Ecosystem.' We'll touch on AI for Capella briefly today, but that session will go much deeper, with an extended demo.

After that, you've probably seen already the dates for Capella Days 2026 edition, our annual free and online conference that gathers hundreds of members of our community, so the conference is happening December 1st through 3rd — the program this year will be announced on September 17th and thanks for all the contributions we received this year again, that's awesome!

And before we start, let me quickly introduce myself for those who don't know me.

I'm Stéphane Lacrampe, one of the founders of Obeo and Managing Director for North and South America. I've been in the Capella adventure since the early days, and I've had the chance to watch Capella grow from a Thales-internal tool into the open-source MBSE reference it is today, I am generally pretty active in the Capella/MBSE community and if you're going to conferences in the field like the INCOSE ones, chances are that you can meet me there.

With that, let's get into it, as I have a pretty packed agenda for today!

Here's a quick look at what we'll cover today. I'll start with a short refresher on Arcadia and Capella for anyone joining us for the first time. Then we'll look at the Capella community — how active and how global it's become. From there, we'll dive into what's new in Capella 7.1.0, and share some exciting news about Obeo's growing commitment to the project. We'll then go through the latest on Capella's open-source add-ons, followed by an update on other commercial add-ons and products, including a first look at AI for Capella. And we'll close with something bigger: Capella for SysON, and what it means for the future of Capella.

Let's get started with the basics.

So, first, for those who have never heard of Arcadia and Capella:

Arcadia is a Model-Based Systems Engineering method — closely aligned with ISO 15288 — that guides systems engineers step by step, from stakeholder needs down to concrete architecture and design choices, while keeping the whole system model consistent.

[Click] Here's a typical representation of a model built using Arcadia. It illustrates core systems engineering principles — like the clear separation between the definition of the need and the definition of the solution, and traceability across the different engineering perspectives.

[Click] And Capella is the free, open-source tool that puts Arcadia into practice.

[CLICK] It lets engineers model, visualize, and document a system's architecture in a very clear

Capella is, above all, a modeling tool — here are a few examples of the kinds of diagrams you'd create with it.

But beyond these diagram examples, I'd say the real strength of Capella comes from how tightly it's coupled with the Arcadia methodology. Capella creators puts a lot of effort into simplifying modeling activities, so that building a consistent, coherent system architecture model is accessible to any systems engineer — not just modeling experts. The result is a tool that's trully comprehensive and rich, but with a smooth learning curve.

Capella isn't just a tool and a method — it's also a vibrant ecosystem, bringing together individuals and organizations around its open-source nature.

[Click] Later in this webinar, I'll walk you through some of the open-source and commercial add-ons that extend Capella's capabilities in different directions. These add-ons are, largely used across our community.

[Click] And finally, it's worth mentioning that our community also includes organizations providing services — training, coaching, integration — helping teams succeed with their MBSE adoption.

All right, with that context in place, let's talk about the community itself.

Over the years, we've built a vibrant and active ecosystem around Capella: more than 80 webinars organized since 2017, and nine editions of Capella Days. I want to take a moment here to thank all of you in the community who take the time to share your experience with Capella — it truly makes a difference.

We make all of this content available on our Capella YouTube channel, which now has more than 180 videos — I think it's fair to say this is probably the most extensive free content library available anywhere for MBSE.

We're also very active on LinkedIn with more than 500 posts a year, and our forum sees about 150 messages a month — a real, living conversation between practitioners.

One more thing worth highlighting on the community side: Capella content is now available in eight languages.

This reflects the work we've been doing on internationalization — making Capella accessible to engineers everywhere, not just in English-speaking or European markets.

I would like to thank here our partner in Thailand, Norasi Team, and more particularly Vorachet, for contributing last month the translation for Thailand.

It's a small thing on paper, but it reflects something bigger: Capella's community and reach keep expanding well beyond its original footprint.

Beyond our own events, Capella has a real presence on international conferences.

We're regulars at INCOSE's International Workshop and International Symposium, and at their regional chapters worldwide — AOSEC in Asia, EMEA in Europe, WSRC in the US, SETE in Australia, and events in Japan, Korea, and India. We're also present at ESA's MBSE conference, IEEE SysON, Siemens Realize Live, CSDM, and AFIS events.

Altogether, that's more than 160 conferences since 2015 — as sponsors, with a Capella booth, giving talks, or running tutorials. It's another way the community stays connected and keeps growing, well beyond what happens online.

And all of that has helped Capella grow to over 1,500 organizations that have or are using Capella today.

And this year alone has been particularly active.

We've already run seven webinars since January. Capella Days is set for December and early talks lined up with Rolls-Royce, Indra Space, and Bluekei.

On the conference side, we've been or will be at the INCOSE's International Symposium in Yokohama, EMEASEC in Sweden, the Dutch SE Days, TdSE in Germany, CSD&M in France, ASEC in the UK, and AeroSEC in India. Not mentioning the local Capella meetups in Australia and in Brazil.

So beyond the technology, it is great to see how this community is very much alive.

Now, let's talk about the technology itself, starting with what's new in Capella 7.1.0.

Capella 7.1.0 was released on August 3rd.

One of the important work we did for this version is strengthening its foundations — a more robust, secure, and easier-to-maintain platform over the long term. Concretely, that means moving from Java 17 to Java 21, a more recent and sustainable runtime. It also means a real cybersecurity push: nine CVEs addressed through dependency updates and removals listed on the slide.

This kind of change is also very much in line with the direction set by the EU Cyber Resilience Act that is impacting the whole software industry.

This groundwork may not be the most visible part of 7.1.0, but it's essential — it gives us a stronger foundation to build the next generations of Capella on. And we'll keep carrying these evolutions forward in upcoming versions.

Next, internationalization. We did a significant amount of work toward making the tool more accessible to international users — the platform now provides the foundations needed to translate Capella's user interface into different languages.

To be clear: we don't ship the translations themselves yet — that part still needs to be provided separately, and the community is welcome to contribute. But our partner Siemens has already leveraged this foundation to provide a Japanese version of System Modeling Workbench for its next release.

Let me show you a very quick [demo] of that.

Now, something I know a lot of you have been waiting for: enhancing diagram layout — probably the number one pain point in Capella.

7.1.0 includes a first integration of the Eclipse Layout Kernel technology, giving you access to a variety of automatic layout algorithms. This is still experimental — we've tested it mainly on architecture diagrams — but you can try it and give us feedback. Here's an example of what you get. First, a typical diagram layoutted by hand. [CLICK] Then an horizontal layout, [CLICK] and a vertical Layout.

[NO] *(Let me show you a quick [DEMO] of that.)*

I've put a link to the forum with details on how to install this optional feature.

A few other improvements made it into this release: project migration through APIs, REC/RPL fixes, and a fix for color display issues on Linux. You'll find the full details in the 7.1.0 release notes, [linked here](#).

And last but not least [CLICK] we have a new splash screen for Capella 7.1.0. But there's a bigger meaning behind that than just a nicer splash screen, which I'll get to in the next slide.

While Thales developed Capella and Arcadia method, Obeo has been part of this journey from the early years. We've contributed to key technologies that support Capella — Eclipse Sirius, EMF Services, GMF Runtime, and other Eclipse technologies — to support Thales along the way. Over time, our involvement has expanded well beyond the platform itself, especially since Capella became open-source in 2015.

We have recently announced at Obeo our decision to strengthen this commitment. Obeo will contribute significantly more to Capella itself going forward. We're moving to a new release cycle, with 3 to 4 releases per year — and adopting a new version numbering scheme, so it's clearer which version you're using and when it was released. That means Obeo is taking a much more significant role in the Capella project and its leadership — ensuring the long-term sustainability of Capella, and better

So what's coming in the next release, planned for November.

- First, we're ramping up our capability to release Capella and all its underlying components, making sure we are able to deliver high quality releases. Behind the scenes, that means electing new committers, and refining our build, testing, and validation procedures.
- Then, we are working on a significant Eclipse Platform upgrade, from 2023-03 to 2025-09, an upgrade across all Capella dependencies, for an improved OS compatibility and reduced CVE exposure
- We will be activating the the new Calendar Version numbering for Capella, so it won't be a 7.2 or 8.0, but a 2026.11
- We will also work on various bug fixes and improvements — including continued work on ELK integration.

But the evolution of Capella isn't just about Capella itself — so let's talk about add-ons.

Let's start with a new open-source add-on: the Capella Simulink Connector, developed by Thales and that we recently helped them to open-source it.

It integrates Capella state machines and functional chains directly with Simulink for simulation.

This capability has been long awaited by the Capella ecosystem, and now it's available as open source.

The installation documentation and a new dedicated webinar presentation are coming soon, but I put a link on screen if you want to check out the recording already available of a webinar previously made by Thales.

Feedback and contributions are very welcome on this new exciting add-on.

This next contribution comes from Rolls-Royce, and it improves system to Subsystem Transition — the add-on that turns a set of components into a proper subsystem in a new Capella project.

The previous implementation assumed the component perimeter never changed once a subsystem was created. When it did change, the transformation could corrupt the model.

The fix does three things: it now shows you the perimeter change upfront in the dialog before applying anything; it matches elements by their role instead of technical ID, so existing elements are updated rather than destroyed and recreated; and as a result, impacts on the rest of the model and its diagrams are much more contained. A related bug in the Cancel button — which could unexpectedly modify the model — was fixed too.

Now, a quick update on M2Doc, our Word document generation add-on.

Since version 4.0.0, M2Doc supports Office 365 natively, and instructions are now plain text, while it was previously in hidden fields.

Since then, we've kept reinforcing the HTML-to-Word integration — lists, tables, images — across several releases. We've also added support for section and page numbering through bookmark references.

And on the extension side, templates can now directly call a Semantic Browser query — for example, this query on the slide applied on a logical function pull exactly what you'd get from the Semantic Browser in Capella straight into your generated document.

A quick update on Python for Capella as well. This add-on is not that old but is now largely used by Capella users.

The big addition here is the new Diagram API. Until now, you could script the model, but not the diagrams themselves. Now you can create diagrams — for example a LAB diagram — copy an existing one, and customize its style, all from a script. That opens the door to automating diagram production, which matters a lot for anyone generating a lot of standardized views. Let me show you a quick demo of that [VIDEO DEMO]

Alongside that, we've fixed various issues and improved packaging. And looking ahead, we've also built an Acceleo-based generator that can produce a Python for Capella API from any Ecore model — which opens the door to easily extending

Next, a new open-source add-on: the Capella Traceable Properties Values Edition, funded by Rolls-Royce.

It's built around an idea called the Holistic Requirements Model: functional requirements describe what the system must do, but that's only half the picture. Non-functional requirements — things like mass, cost, performance, efficiency — describe how well the system does it. You need both to really capture what's expected of a system.

Capella already has the right building blocks for this: Property Values and Property Value Groups. This add-on doesn't introduce anything new into the metamodel — it reuses those existing concepts, along with Generic Trace links, and makes them much easier to work with, through new palette tools and dedicated diagram types. Concretely: you can apply property values directly on your existing

All right, now let's look at some of the main updates in some of the commercial add-ons and products around Capella.

Let's start with Team for Capella, our real time collaboration add-on

Looking back at the last couple of years, the biggest thread has been authentication and security, including a complete rework of OpenID Connect support

Alongside that, we made a real performance leap on the Semantic Browser, refresh times on large connected projects went from close to a minute of freeze down to under a second.

And underneath, we fixed a memory leak in the CDO revision cache and kept up continuous CVE cleanup on Jetty and Jenkins.

Now, with the version 7.1.0. which was just officially released this week, the headline change: secure connections are the default now when installing T4C: WSS and HTTPS are out of the box, while unencrypted TCP or HTTP are explicitly

A word on Publication for Capella, our web publication and traceability solution.

Over the past year, we've extended our ALM coverage with Jama Connect, and Jira Cloud is now supported alongside Data Center thanks to SodusWillert OSLC add-on update

The 2026.5 release focused on smoother administration at scale: a new 'Organizations' concept, a dedicated Server Admin role, OIDC team sync, and a centralized dashboard for licenses and roles. We also laid the foundations for real version control, and diagram images now publishable as Polarion work items.

Right now, we're bringing CodeBeamer up to full traceability parity with our other ALM integrations thanks to a great collaboration with PTC, adding decorators in the online explorer, and exposing

A word on System Modeling Workbench, our joint development with Siemens.

SMW 2606 moved to Capella 7.1 and Java 21, with most effort going into security and robustness.

Next up in 2612: cleaning up more CVEs, rich-text images support and SMW subsystem reuse from Teamcenter, as well as multiple language support as mentioned earlier in the presentation.

One important thing to mention though: we're packaging Siemens EDA's VSE Analysis add-on into SMW, letting HW architects model electronics — chips, boards, network topology — directly in Capella, then simulate it through Siemens' Vista engine. It's a strong example of Capella reaching beyond systems architecture into hardware and chip-level verification.

And now, something new and exciting: AI for Capella. I won't go deep today — we have a dedicated webinar on this at the end of September — but I want to give you the shape of it.

The idea isn't to bolt one built-in AI assistant onto Capella. It's to make Capella an agent-ready platform — so it can connect to whatever AI ecosystem your organization already uses, whether that's Claude, Codex, Mistral, or a locally hosted model.

Architecturally, Capella exposes a controlled set of capabilities through an MCP server. An external AI agent connects through that layer, and from there it can also reach your other tools and data sources as it runs in your typical prompting local tool. Capella itself remains where the model is actually read and modified, leveraging the whole platform.

And there are many other open-source or commercial add-ons in the ecosystem, I obviously cannot cover them all today, but we have organized webinars on topics like EcoDesign for Capella or ATICA in the last months, they are all available on the Capella Youtube channel if you're interested in a specific one.

Now, let's talk about something I am really excited about: Capella for SysON.

For more than a decade, Capella and the Arcadia method have helped systems engineers tackle the complexity of designing large, critical systems. But the systems engineering landscape is entering a new phase — shaped by major shifts like the emergence of SysML v2 as a new industry standard, AI of course and the growing move toward web-based engineering environments.

At Obeo, we believe the principles that made Capella and Arcadia successful are still highly relevant in this new landscape. That's why we're launching Capella for SysON — a new open-source initiative that progressively builds the convergence of Capella, Arcadia, SysML v2, and web-based modeling.

Our long-term ambition is for Capella for SysON to become the next generation of Capella. Today, I

Let's start with SysML v2 itself, since it's central to everything we'll cover next. Obeo is a member of the OMG, we're part of the SysML v2 Submission Team, and we're actively contribute to the definition of the SysML v2 standard.

SysML v2 is the next-generation systems modeling language, formally approved by the OMG in 2025. It's built on much stronger semantic foundations — KerML — which brings real improvements in precision, expressiveness, and consistency compared to SysML v1. It also comes with a native textual notation alongside the graphical one, and a standardized API. That last point matters a lot: SysML v2 is positioning itself as the future interoperability standard across the MBSE tool ecosystem.

But SysML v2 also comes with a real challenge. It is a rich, powerful, general-purpose language —

Before introducing Capella for SysON itself, let me quickly walk through the two building blocks it stands on.

The first is Sirius Web : a low-code modeling platform, some principles as Eclipse Sirius but in a browser. We've been investing on this new platform since 2019, and now this technology is already deployed and used throughout our customer base. The point is that Capella itself is built on Sirius, the desktop predecessor. So in a sense, it is quite natural that we are building the future of Capella on top of Sirius Web.

The second is SysON — the leading open-source, web-based tool for editing SysML v2 models, built on top of Sirius Web. We are co-developing it with the CEA, and it's hosted at the Eclipse Foundation. What makes SysON particularly relevant here is that it's flexible by design: it works out of the box for

So what exactly is Capella for SysON?

At its core, it brings the Arcadia approach into SysON. The goal is to let engineers model according to Arcadia — the same concepts, the same engineering approach they already know from Capella — but have those models natively expressed in SysML v2.

Beyond Sirius Web and SysON, Capella for SysON relies on three complementary open-source building blocks.

First, the Arcadia SysML v2 Library, which formalizes Arcadia's core concepts on top of SysML v2

Second, Capella for SysON itself, which brings the User Interface and business logic of the Arcadia methodology into SysON, with Arcadia-specific and Capella-inspired graphical editors.

Let's zoom in on the first building block: the Arcadia SysML v2 Library.

The design of this library is driven by one main priority: interoperability. Our goal is, for models built with this library, to look as close as possible to what you'd naturally produce in SysML v2 without it. In other words, not an extensive, Arcadia-specific library that maps one-to-one to the Capella metamodel and produces heavily 'Arcadia-colored' models that only make sense to Arcadia tools. The whole point of using SysML v2 is to stay interoperable with the broader ecosystem — simulation, analysis, verification tools — and that's what we are aiming at.

At the same time, we want these models to remain easy to understand and manipulate for Arcadia users. So we've struck a balance: the library provides a minimal but essential semantic layer —

Now, let's actually look at Capella for SysON.

What you're about to see is a graphical editor that looks and feels like Capella — the same kind of diagram, the same vocabulary — except every element on screen is a SysML v2 element, specialized using the Arcadia library we just covered. So underneath the familiar diagram, you're really looking at a standard SysML v2 model.

Let me show you a quick, early demo — this is very much a work in progress, so bear with the rough edges...

[Démo]

As you can see, this is still early — but the core idea is there: Arcadia's modeling experience, delivered natively in a web-based SysML v2 environment.

So where is all of this heading? Our ambition at Obeo goes beyond just providing an Arcadia extension for SysON.

We intend Capella for SysON to progressively become the next generation of Capella with the following pillars:

- an environment that's web-based, for easier deployment and collaboration;
- fully based on SysML v2;
- built for collaboration, leveraging SysON's architecture;
- and grounded in Arcadia, preserving the methodological approach that's central to everything Capella does.

But let's be clear about what this isn't: it's not an overnight replacement. Capella is a mature engineering environment, used on complex, long-running projects, with hundreds of wizards, helpers,

So here's our current roadmap.

2026 is about building the open-source foundations: the Arcadia SysML v2 Library, the Capella SysML v2 Bridge, and Capella for SysON itself — plus, on the enterprise side, Obeo Enterprise for SysON, which brings the enterprise-grade capabilities needed for industrial deployment down the line.

In 2027, our objective is a first MVP by end of June 2027.

And after that, we complete the vision — broader coverage of Arcadia and Capella capabilities on SysML v2, continued improvements to interoperability and the migration path, and ongoing innovation around collaboration, supervision, and AI for industrial use.

Now let's talk about the Capella SysML v2 Bridge.

What it does is transform existing Capella Desktop models in SysML v2.

This serves two purposes. First, it's a genuine migration path: a way to progressively move toward SysML v2 and, eventually, Capella for SysON, without a disruptive cutover. Second, it gives you interoperability today: your Capella models can be connected to the broader SysML v2 ecosystem, simulation tools, analysis tools, and so on, well before Capella for SysON itself is mature enough for production use.

And this transformation is more natural than you might expect: what's changed is that v2 introduced much richer language constructs, like much better structural-behavioral integration that simply didn't exist in v1. Those constructs happen to give us clean, natural ways to express Arcadia concepts

To conclude on this topic, I want to insist on the fact that this roadmap does not mean we're stepping back from Capella. Quite the opposite. As we covered earlier today, Obeo is increasing its investment in Capella. Capella Desktop is not a legacy platform we're planning to phase out; it's a vibrant and growing community and a platform we're actively evolving for the long term. Capella for SysON is an additional bet on the future, not a signal that we're pulling back from the present.

Capella Desktop remains the mature, proven solution for operational Arcadia projects, and Capella for SysON is still at an early stage.

There's still significant work ahead, and we want to build this future through Open Innovation, together with organizations that see value in bringing Arcadia and Capella into the SysML v2 and web era. If this direction resonates with your own MBSE

Before we open it up for questions, let me pull the threads together.

First, Capella 7.1.0 is stronger, more secure, and more international — and as I mentioned, it's really just the beginning of something bigger.

Second, that 'something bigger' means Obeo is investing more in Capella: more releases per year, new committers, deeper contributions.

Third, the ecosystem around Capella keeps growing — new add-ons, real contributions from industrial partners like Rolls-Royce and Thales, and a community that's genuinely active worldwide.

And fourth, Capella for SysON gives you a stable, long-term path forward as SysML v2 becomes part of the landscape — not a disruption to plan around, but a natural evolution built on the same Arcadia

First, a quick heads-up on what's coming right after this.

On September 24th, I'll be hosting another webinar: 'Connect Capella to Your AI Ecosystem.' We'll touch on AI for Capella briefly today, but that session will go much deeper, with an extended demo.

Save the date as well for Capella Days 2026, online, December 1st through 3rd — the program will be announced on September 17th, so keep an eye out.

And if you or your team want to go further with Arcadia and Capella, we have training sessions running through the rest of the year — dates and registration link are up on screen.

